dielectric-IDEs: Quick-Start Guide

Python code accompanying J. S. Colton et al., "Measuring Dielectric Functions of Thin Films Using Interdigitated Electrodes, with Application to Metal Halide Perovskites" (2026). Code and data: https://github.com/coltonlab/dielectric-IDEs. The examples below use the data files of the paper, which are in the data/ folder, so you can run them and compare your output with what is printed here. The code and data are released under the MIT License (see LICENSE).

1. Before you start

- Install Python 3 with numpy and scipy (matplotlib only for plots): pip install -r requirements.txt
- Measure, versus frequency, a blank IDE (bare substrate) and an IDE of the same design with your film on it.
- Measure the film thickness and the electrode dimensions (gap s, finger width w, electrode thickness t_e, number of fingers N, overlap length L). The film thickness dominates the uncertainty.
- Run every command from the dielectric-IDEs folder. Each script lists its options with --help and begins with a comment block explaining what it does.

2. Which script do I use?

I want to…	Run	Main input
calculate the dielectric constant of a film	extract_permittivity.py	blank and film data files, film thickness
estimate its uncertainty	uncertainty.py	same as above
get the imaginary part ε_i (Kramers-Kronig)	kramers_kronig.py	the CSV file written by extract_permittivity.py
check the model against a finite-element calculation (2D)	fem2d.py	dielectric constants, film thickness, electrode dimensions
check the finger-end correction N_eff (3D, advanced)	fem3d_end_correction.py	electrode and chip dimensions, film
reproduce the numbers in the paper	reproduce_paper.py	the data/ folder (and --paper options of the other scripts)

3. Data file format

Plain-text CSV, one row per frequency, no header needed. The data files of the paper are named `<compound>_<sample>.csv` (for example `MAPbI3_s1.csv`), plus `blank_main_batch.csv`, `blank_second_batch.csv` and `water_1cm.csv`; `data/samples.csv` gives each file's sample, compound, film thickness, wafer batch and blank, and its original name in our lab records. Our files have four columns (HP 4192A analyzer, parallel mode):

Column	Quantity	Unit
0	frequency f	Hz
1	admittance magnitude $\lvert Y \rvert$ (not used)	S
2	parallel conductance G	S
3	parallel capacitance Cp	F

First lines of data/MAPbI3_s1.csv:

```
5.000000000000001,0.000000214250000,0.000000214500000,0.000000000188750
5.795884707147059,0.000000215375000,0.000000215500000,0.000000000165000
```

If your columns are in a different order, give their numbers (counting from 0) as --columns f,G,Cp, for example --columns 0,2,1. If you have no conductance, use −1 for G (--columns 0,-1,1); the loss correction is then skipped.

4. Example: dielectric constant of a film

$MAPbI_3$ film s1 (437.8 nm thick) on our IDEs, with the blank of the same wafer batch:

```
python extract_permittivity.py --blank data/blank_main_batch.csv \
    --film data/MAPbI3_s1.csv --thickness-nm 437.8 \
    --out MAPI_s1.csv --plot
```

It takes about 20 seconds and prints (spaces shortened and long lines wrapped here):

```
N_eff = 130.892;  results written to MAPI_s1.csv
  f =    1019.4 Hz  eps_substrate = 4.5791  film eps_r = 95.3
                    (Cp only 105; eps'' from G 543)
  f =   99312.1 Hz  eps_substrate = 4.5346  film eps_r = 48.83
                    (Cp only 48.91; eps'' from G 8.58)
  f = 1055353.3 Hz  eps_substrate = 4.5319  film eps_r = 46.93
                    (Cp only 46.93; eps'' from G -0.171)
  f = 9674842.1 Hz  eps_substrate = 4.5179  film eps_r = 46.25
```

The value at 9.67 MHz, 46.25, is the 46.2 of Fig. 7 of the paper. MAPI_s1.csv has one row per frequency (100 Hz–10 MHz by default) with these columns:

- eps_substrate: the substrate (quartz) dielectric constant from the blank.
- eps_r_Cp_only: the film's ε_r from the measured capacitance alone.

- eps_r_loss_corrected, eps_loss_incl_conduction: ε_r and ε'' from fitting the complex capacitance $C_p - iG/\omega$, below 2 MHz. Fitting C_p alone overestimates ε_r of a lossy film. ε'' includes DC conduction and any error in G (here it is negative near 1 MHz, where our G had a setup error).
- eps_r: the recommended value, as in the paper (loss-corrected below 2 MHz, Cp only above).

For your own electrodes add --s-um 10 --w-um 10 --te-um 0.1 --N 50 --L-mm 2.0 (gap, width and thickness in µm, number of fingers, overlap in mm). Other useful options: --fmin/--fmax (frequency range), --loss-cutoff (where your G is reliable), --bulk (a liquid such as water covering the electrodes), and no --film at all to analyze only the substrate.

5. Example: uncertainty

```
python uncertainty.py --blank data/blank_main_batch.csv \
    --film data/MAPbI3_s1.csv --thickness-nm 437.8 \
    --out MAPI_s1_unc.csv
```

Last line printed (9.67 MHz; wrapped here), in percent of ε_r:

```
f = 9674842.1 Hz  eps_r = 46.25  total 6.50%  (thickness_abs 4.42,
    thickness_film_to_film 4.26, blank_chip_to_chip 1.54, hf_instrument 1.39,
    geometry_model 0.50, resolution 0.10)
```

This is the 46.2 ± 3.0 of the paper. The six items are those of Sec. IVB: absolute thickness (±20 nm), film-to-film thickness variation (4.4%), chip-to-chip variation of the blank (0.3%), the analyzer's error at high frequency (0.27% at 9.67 MHz, $\propto f^2$), electrode geometry and model (0.5%), and capacitance resolution (0.0041 pF). These defaults describe our measurements; set your own, e.g. --dt-nm 10 if your thickness is known to ±10 nm. MAPI_s1_unc.csv gives every item at every frequency.

6. Example: imaginary part by Kramers-Kronig

```
python kramers_kronig.py --input MAPI_s1.csv --plot
results written to MAPI_s1_KK.csv; fit rms 0.264%
eps_i at 1.055e+06 Hz = 1.416 (NOT determined)
```

The fit gives ε_i at every frequency, but near the ends of the measured range ε_i is not fixed by the data. The column eps_i_determined is 1 where ε_i changes by less than 20% when the smoothing is changed by a factor of 30 and when the range is shortened by two points at each end (Appendix B). For this film, $\varepsilon_i = 5.16$ at 63.8 kHz is determined (variants 5.11–5.16); $\varepsilon_i = 1.40$ at 1.42 MHz is not (variants 0.68–1.75). For thick films with large low-frequency signals the paper extends the input down to 5 Hz: run extract_permittivity.py with --fmin 4 --substrate-fixed-below 90 first.

7. Example: finite-element checks

2D check of the model for a given film and geometry (fem2d.py, about 10 s):

```
python fem2d.py --eps-film 47 --eps-sub 4.52 --thickness-nm 437.8
FEM (30240 nodes): S = 7.1796289   model: S = 7.1787841
    model/FEM - 1 = -0.0118%
```

S is the capacitance in the units of the model ($C = \varepsilon_0 N_{eff} L (\pi/8) S$). The finite-element value decreases toward the true capacitance as the mesh is refined; with --refine 64 --growth 0.04 it falls below the model, which then lies 0.008% above it, as in the paper. Use this to check the model for thick electrodes (large t_e/a), where its error grows (about 0.6% at $t_e/a = 0.06$).

3D finger-end correction (fem3d_end_correction.py, advanced, about 10 s per run):

```
python fem3d_end_correction.py
bare substrate: the ends add 1.678% to the mutual capacitance
    (0.78 a of extra overlap per end)
```

Add --thickness-nm and --eps-film for a film: the script then reports the extra overlap per finger end for the film's contribution, which the rule $N_{eff} = N - ½ + N\, a/L$ takes as 0.5a (0.54a–0.57a for our films). Change --a-um, --w-um, --L-um, --gap-um, --B-um and --tq-um for your chip.

8. Reproducing the paper

Command	Reproduces	Time
python reproduce_paper.py	all ε_r values in the paper (prints "18 of 18 values agree")	≈ 4 min
python uncertainty.py --paper	the uncertainties of Sec. IVB	≈ 1–2 min
python kramers_kronig.py --paper	ε_i of all films (Fig. 5, Sec. IVB)	≈ 7 min
python fem2d.py --paper	Supplemental S4 and S9; recovery test of Sec. IIIB6	≈ 3 min
python fem3d_end_correction.py --paper	Appendix C (1.68%; 0.54a–0.57a per end)	≈ 2 min
python make_si_data_files.py	the Supplemental data files S6_data_*.csv (run the first and third rows first)	≈ 2 min
python check_model.py	quick self-test (conformal mapping, convergence)	seconds

9. Using the model in your own Python code

```
from ide_model import IDEGeometry, IDEModel
geom = IDEGeometry(s_um=50.27, w_um=29.78, te_um=0.12, N=130, L_mm=7.475)
model = IDEModel(geom)
# measured capacitances in farads: blank, then film
eps_Q = model.solve_substrate(19.887e-12)
eps_P = model.solve_film(24.424e-12, eps_Q, t_film_um=0.438)
print(f"{eps_Q:.3f} {eps_P:.2f}")     # prints 4.520 47.01
```

The model also accepts complex dielectric constants (solve_substrate_complex, solve_film_complex). The comments at the top of ide_model.py explain the geometry, the method and the units.